

ADVANCED

Stage 6

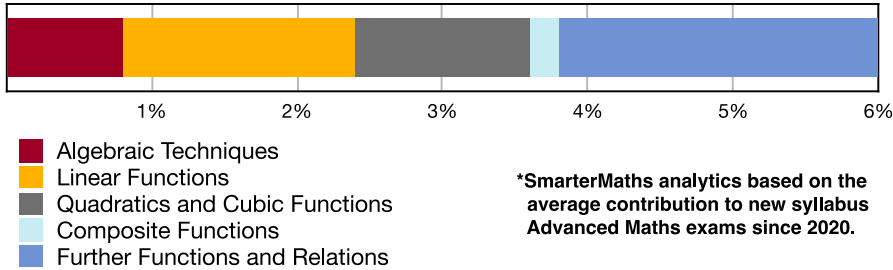
Functions (Adv), F1 Working with Functions (Adv)

Further Functions and Relations (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 109.5 minutes (based on allocation of 1.5 minutes per mark)

F1 Working With Functions



HISTORICAL CONTRIBUTION

- F1 Working with Functions* is classic low hanging fruit, contributing a healthy 6.0% per exam since the new Advanced syllabus was introduced in 2020.
- We have split the topic into 5 categories for analysis purposes: 1-*Algebraic Techniques* (0.8%), 2-*Linear Functions* (1.6%), 3-*Quadratics and Cubic Functions* (1.2%), 4-*Composite Functions* (0.2%) and 5-*Further Functions and Relations* (2.2%).
- This analysis looks at the largest sub-topic, *Further Functions and Relations*.

HSC ANALYSIS - What to expect and common pitfalls

- Further Functions and Relations* (2.2%) was most recently tested in multiple-choice questions in 2023-24, and in a chunky 4-mark question in 2022.
- Proportional relationships was examined as an *Adv/Std2* common question in 2022 *Adv 12* and surprisingly produced a sub-50% mean mark in part (a) which should be reviewed.
- Absolute value* equations can be examined in the new course, although it must be emphasised that *absolute value* inequality equations have moved to the *Ext1* course.
- Graphing absolute value* equations should be a revision focus in our view, as we expect this skill to be examined more frequently going forward.
- Reflections require students to graph $y = -f(x)$ and $y = f(-x)$ of given equations. Examined in the first new syllabus paper (2020 *Adv 24*) within the context of a circle function, this question and other reflection examples in the database warrant attention.

Questions

1. Functions, 2ADV F1 2019 HSC 2 MC

What values of x satisfy $4 - 3x \leq 12$?

- A. $x \leq -\frac{16}{3}$
- B. $x \geq -\frac{16}{3}$
- C. $x \leq -\frac{8}{3}$
- D. $x \geq -\frac{8}{3}$

2. Functions, 2ADV F1 2013 HSC 3 MC

Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+3}}$?

- A. $x > -3$
- B. $x \geq -3$
- C. $x < -3$
- D. $x \leq -3$

3. Functions, 2ADV F1 2020 HSC 1 MC

Which inequality gives the domain of $y = \sqrt{2x-3}$?

- A. $x < \frac{3}{2}$
- B. $x > \frac{3}{2}$
- C. $x \leq \frac{3}{2}$
- D. $x \geq \frac{3}{2}$

4. Functions, 2ADV F1 2023 HSC 3 MC

What is the domain of $f(x) = \frac{1}{\sqrt{1-x}}$?

- A. $x < 1$
- B. $x \leq 1$
- C. $x > 1$
- D. $x \geq 1$

5. Functions, 2ADV F1 2024 HSC 6 MC

What is the domain of the function $f(x) = \frac{1}{\sqrt{x^2 - 1}}$?

- A. $[-1, 1]$
- B. $(-\infty, -1] \cup [1, \infty)$
- C. $(-1, 1)$
- D. $(-\infty, -1) \cup (1, \infty)$

6. Functions, 2ADV F1 EQ-Bank 24 MC

Given that $f(x) = \begin{cases} 3 - (x - 2)^2, & \text{for } x \leq 2 \\ mx + 5, & \text{for } x > 2 \end{cases}$

What is the value of m for which $f(x)$ is continuous at $x = 2$?

- A. 1
- B. 2
- C. -1
- D. -2

7. Functions, 2ADV F1 SM-Bank 21 MC

A circle with centre $(a, -2)$ and radius 5 units has equation

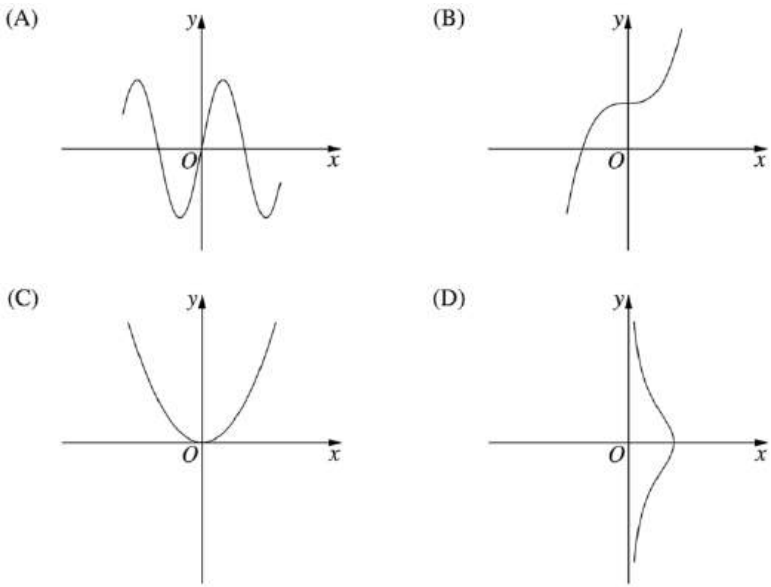
$x^2 - 6x + y^2 + 4y = b$ where a and b are real constants.

The values of a and b are respectively

- A. -3 and 38
- B. 3 and 12
- C. -3 and -8
- D. 3 and 18

8. Functions, 2ADV F1 2016 HSC 4 MC

Which diagram shows the graph of an odd function?



9. Functions, 2ADV F1 2023 HSC 9 MC

Let $f(x)$ be any function with domain all real numbers.

Which of the following is an even function, regardless of the choice of $f(x)$?

- A. $2f(x)$
- B. $f(f(x))$
- C. $(f(-x))^2$
- D. $f(x)f(-x)$

10. Functions, 2ADV F1 2007 HSC 1b

Solve $2x - 5 > -3$ and graph the solution on a number line. (2 marks)

11. Functions, 2ADV F1 2009 HSC 1c

Solve $|x + 1| = 5$. (2 marks)

12. Functions, 2ADV F1 2010 HSC 1d

Solve $|2x + 3| = 9$. (2 marks)

13. Functions, 2ADV F1 2006 HSC 1e

Solve $3 - 5x \leq 2$. *(2 marks)*

14. Functions, 2ADV F1 2011 HSC 1e

Solve $2 - 3x \leq 8$. *(2 marks)*

15. Functions, 2ADV F1 2017 HSC 11g

Solve $|3x - 1| = 2$. *(2 marks)*

16. Functions, 2ADV F1 2018 HSC 11b

Solve $1 - 3x > 10$. *(2 marks)*

17. Functions, 2ADV F1 2008 HSC 1d

Solve $|4x - 3| = 7$. *(2 marks)*

18. Functions, 2ADV F1 SM-Bank 33

- i. State the domain and range of $y = -\sqrt{12 - x^2}$. *(2 marks)*
 - ii. Sketch the graph. *(1 mark)*
-

19. Functions, 2ADV F1 SM-Bank 41

Find the values of x for which $|x + 1| = 5$. *(2 marks)*

20. Functions, 2ADV F1 SM-Bank 42

Find the values of x for which $|x - 3| = 1$. *(2 marks)*

21. Functions, 2ADV F1 SM-Bank 44

Solve $|x - 2| = 3$. *(2 marks)*

22. Functions, 2ADV F1 SM-Bank 37

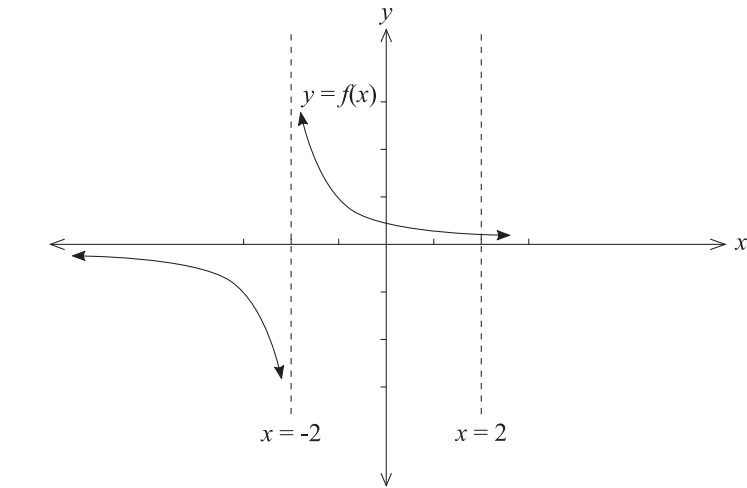
Find all values of x for which $|x - 4| = \frac{x}{2} + 7$. *(3 marks)*

23. Functions, 2ADV F1 2016 HSC 11a

Sketch the graph of $(x - 3)^2 + (y + 2)^2 = 4$. *(2 marks)*

24. Functions, 2ADV F1 SM-Bank 36

Consider the function $f(x) = \frac{1}{x + 2}$.



- i. Sketch the graph $y = f(-x)$. *(2 marks)*
 - ii. On the same graph, sketch $y = -f(x)$. *(2 marks)*
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25. Functions, 2ADV F1 2019 HSC 13e

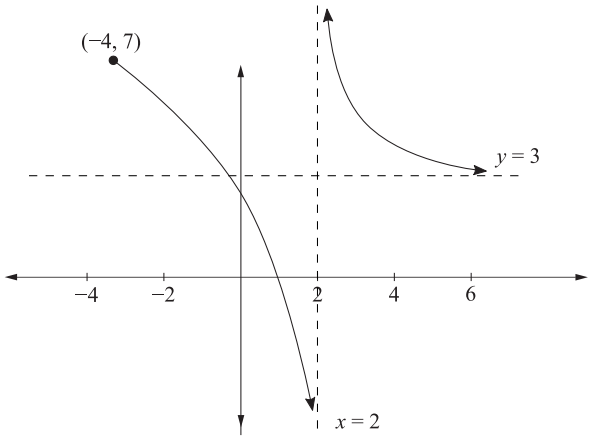
- i. Sketch the graph of $y = |x - 1|$ for $-4 \leq x \leq 4$. *(1 mark)*
 - ii. Using the sketch from part i, or otherwise, solve $|x - 1| = 2x + 4$. *(2 marks)*
-

26. Functions, 2ADV F1 EQ-Bank 45

Determine whether the function $f(x) = 2x^3 - 5x$ is even, odd or neither. Show all working. *(2 marks)*

27. Functions, 2ADV F1 EQ-Bank 6

The graph of $f(x)$ is shown below. It has asymptotes at $y = 3$ and $x = 2$.



Using interval notation, state the domain and range of $f(x)$. (2 marks)

28. Functions, 2ADV F1 EQ-Bank 7

The current of an electrical circuit, measured in amps (A), varies inversely with its resistance, measured in ohms (R).

When the resistance of a circuit is 28 ohms, the current is 3 amps.

What is the current when the resistance is 8 ohms? (2 marks)

29. Functions, 2ADV F1 EQ-Bank 8

Jacques is a marine biologist and finds that the mass of a crab is directly proportional to the cube of the diameter of its shell.

If a crab with a shell diameter of 15 cm weighs 680 grams, what will be the diameter of a crab that weighs 1.1 kilograms? Give your answer to 1 decimal place. (2 marks)

30. Functions, 2ADV F1 EQ-Bank 26

Fuifui finds that for Giant moray eels, the mass of an eel is directly proportional to the cube of its length.

An eel of this species has a length of 25 cm and a mass of 4350 grams.

What is the expected length of a Giant moray eel with a mass of 6.2 kg? Give your answer to one decimal place. (3 marks)

31. Functions, 2ADV F1 EQ-Bank 27

The stopping distance of a car on a certain road, once the brakes are applied, is directly proportional to the square of the speed of the car when the brakes are first applied.

A car travelling at 70 km/h takes 58.8 metres to stop.

How far does it take to stop if it is travelling at 105 km/h? (3 marks)

32. Functions, 2ADV F1 SM-Bank 32

Find the centre and radius of the circle with the equation

$$x^2 - 12x + y^2 + 2y - 12 = 0 \text{ (2 marks)}$$

33. Functions, 2ADV F1 2010 HSC 1c

Write down the equation of the circle with centre $(-1, 2)$ and radius 5. (1 mark)

34. Functions, 2ADV F1 2010 HSC 1g

Let $f(x) = \sqrt{x - 8}$. What is the domain of $f(x)$? (1 mark)

35. Functions, 2ADV F1 2017 HSC 11h

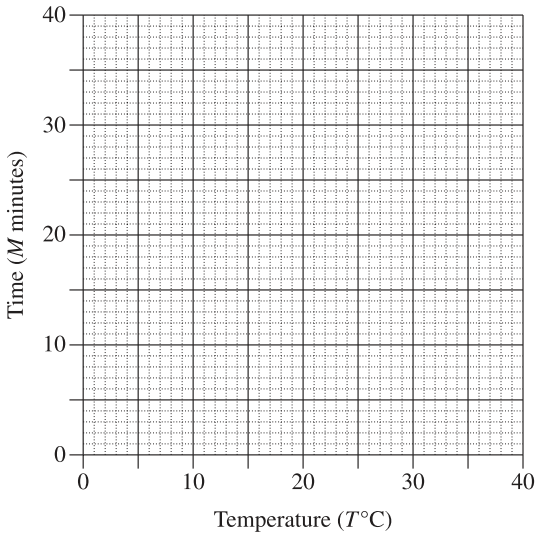
Find the domain of the function $f(x) = \sqrt{3 - x}$. (2 marks)

36. Functions, 2ADV F1 2022 HSC 12

A student believes that the time it takes for an ice cube to melt (M minutes) varies inversely with the room temperature ($T^{\circ}\text{C}$). The student observes that at a room temperature of 15°C it takes 12 minutes for an ice cube to melt.

- a. Find the equation relating M and T . (2 marks)
- b. By first completing this table of values, graph the relationship between temperature and time from $T = 5^{\circ}\text{C}$ to $T = 30^{\circ}\text{C}$. (2 marks)

T	5	15	30
M			



37. Functions, 2ADV F1 2020 HSC 24

The circle of $x^2 - 6x + y^2 + 4y - 3 = 0$ is reflected in the x -axis.

Sketch the reflected circle, showing the coordinates of the centre and the radius. (3 marks)

Worked Solutions

1. Functions, 2ADV F1 2019 HSC 2 MC

$$\begin{aligned} 4 - 3x &\leq 12 \\ -3x &\leq 8 \\ x &\geq -\frac{8}{3} \end{aligned}$$

$\Rightarrow D$

2. Functions, 2ADV F1 2013 HSC 3 MC

Given $f(x) = \frac{1}{\sqrt{x+3}}$

$$\begin{aligned} (x+3) &> 0 \\ x &> -3 \end{aligned}$$

$\therefore$ The domain of $f(x)$ is $f(x) > -3$

$\Rightarrow A$

3. Functions, 2ADV F1 2020 HSC 1 MC

Domain exists when:

$$\begin{aligned} 2x - 3 &\geq 0 \\ 2x &\geq 3 \\ x &\geq \frac{3}{2} \end{aligned}$$

$\Rightarrow D$

4. Functions, 2ADV F1 2023 HSC 3 MC

Cannot have square root of a negative number.

Denominator $\neq 0$

$$\begin{aligned} 1 - x &> 0 \\ -x &> -1 \\ x &< 1 \end{aligned}$$

$\Rightarrow A$

5. Functions, 2ADV F1 2024 HSC 6 MC

$$x^2 - 1 > 0$$
$$x^2 > 1$$

$$x < -1 \cup x > 1$$
$$x \in (-\infty, -1) \cup (1, \infty)$$
$$\Rightarrow D$$

6. Functions, 2ADV F1 EQ-Bank 24 MC

If continuous at $x = 2$:

$$3 - (x - 2)^2 = mx + 5$$
$$3 - (2 - 2)^2 = 2m + 5$$
$$2m = -2$$
$$m = -1$$

$$\Rightarrow C$$

7. Functions, 2ADV F1 SM-Bank 21 MC

$$x^2 - 6x + y^2 + 4y = b$$
Completing the squares:

$$x^2 - 6x + 3^2 - 9 + y^2 + 4y + 2^2 - 4 = b$$
$$(x - 3)^2 + (y + 2)^2 - 13 = b$$
$$(x - 3)^2 + (y + 2)^2 = b + 13$$

$$\therefore a = 3$$
$$\therefore b + 13 = 25 \Rightarrow b = 12$$
$$\Rightarrow B$$

8. Functions, 2ADV F1 2016 HSC 4 MC

Odd functions occur when:

$$f(x) = -f(x)$$

Graphically, this occurs when a function has symmetry when rotated 180° about the origin.

$$\Rightarrow A$$

◆ Mean mark 38%.

9. Functions, 2ADV F1 2023 HSC 9 MC

Even function $\rightarrow f(x) = f(-x)$

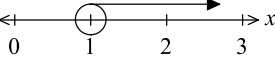
Consider the function $f(x) = x - 2$

$$2f(1) = -2, \quad 2f(-1) = -6 \text{ (not even)}$$
$$f(f(1)) = f(-1) = -3, \quad f(f(-1)) = f(-3) = -5 \text{ (not even)}$$
$$(f(-1))^2 = (-3)^2 = 9, \quad (f(1))^2 = (-1)^2 = 1 \text{ (not even)}$$
$$f(1)f(-1) = -1 \times -3 = 3, \quad f(-1)f(1) = -3 \times -1 = 3 \text{ (possibly even)}$$
$$\Rightarrow D$$

◆◆◆ Mean mark 24%.

10. Functions, 2ADV F1 2007 HSC 1b

$$2x - 5 > -3$$
$$2x > 2$$
$$x > 1$$



11. Functions, 2ADV F1 2009 HSC 1c

$$|x + 1| = 5$$
$$(x + 1) = 5 \quad -(x + 1) = 5$$
$$x = 4 \quad -x - 1 = 5$$
$$x = -6$$

$$\therefore x = 4 \text{ or } -6$$

12. Functions, 2ADV F1 2010 HSC 1d

$$|2x + 3| = 9$$
$$2x + 3 = 9 \quad -(2x + 3) = 9$$
$$2x = 6 \quad -2x - 3 = 9$$
$$x = 3 \quad -2x = 12$$
$$x = -6$$

$$\therefore x = 3 \text{ or } -6$$

13. Functions, 2ADV F1 2006 HSC 1e

$$3 - 5x \leq 2$$
$$-5x \leq -1$$
$$x \geq \frac{1}{5}$$

14. Functions, 2ADV F1 2011 HSC 1e

$$\begin{aligned} 2 - 3x &\leq 8 \\ -3x &\leq 6 \\ x &\geq -\frac{6}{3} \\ x &\geq -2 \end{aligned}$$

15. Functions, 2ADV F1 2017 HSC 11g

$$\begin{aligned} |3x-1| &= 2 \\ 3x-1 &= 2 \quad \text{or} \quad -(3x-1) = 2 \\ x &= 1 \quad \quad \quad -3x = 1 \\ x &= -\frac{1}{3} \\ \therefore x &= 1 \text{ or } -\frac{1}{3} \end{aligned}$$

MARKER'S COMMENT: Note that both conditions must be satisfied! Dealing with negative signs and division for inequalities produced many errors.

16. Functions, 2ADV F1 2018 HSC 11b

$$\begin{aligned} 1-3x &> 10 \\ -3x &> 9 \\ x &< -3 \end{aligned}$$

17. Functions, 2ADV F1 2008 HSC 1d

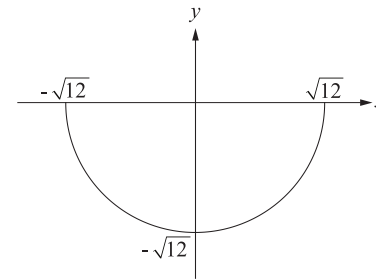
$$\begin{aligned} |4x-3| &= 7 \\ 4x-3 &= 7 \quad \quad - (4x-3) = 7 \\ 4x &= 10 \quad \quad -4x + 3 = 7 \\ x &= \frac{5}{2} \quad \quad \quad -4x = 4 \\ x &= -1 \end{aligned}$$

$$\therefore x = \frac{5}{2} \text{ or } -1$$

18. Functions, 2ADV F1 SM-Bank 33

i. $y = -\sqrt{12-x^2}$
Domain: $-\sqrt{12} \leq x \leq \sqrt{12}$
Range: $-\sqrt{12} \leq y \leq 0$

ii.



19. Functions, 2ADV F1 SM-Bank 41

$$\begin{aligned} |x+1| &= 5 \\ (x+1)^2 &= 5^2 \\ x^2 + 2x + 1 &= 25 \\ x^2 + 2x - 24 &= 0 \\ (x+6)(x-4) &= 0 \\ \therefore x &= 4 \text{ or } -6 \end{aligned}$$

20. Functions, 2ADV F1 SM-Bank 42

$$\begin{aligned} |x-3| &= 1 \\ \text{Method 1} \\ (x-3)^2 &= 1 \\ x^2 - 6x + 9 &= 1 \\ x^2 - 6x + 8 &= 0 \\ (x-4)(x-2) &= 0 \\ \therefore x &= 2 \text{ or } 4 \end{aligned}$$

$$\begin{aligned} \text{Method 2} \\ (x-3) &= 1 \quad \quad -(x-3) = 1 \\ x &= 4 \quad \quad \quad -x + 3 = 1 \\ x &= -2 \end{aligned}$$

21. Functions, 2ADV F1 SM-Bank 44

$$|x-2| = 3$$

$$(x-2)^2 = 3^2$$

$$(x^2-4x+4) = 9$$

$$x^2-4x-5 = 0$$

$$(x-5)(x+1) = 0$$

$$\therefore x = -1 \text{ or } 5$$

22. Functions, 2ADV F1 SM-Bank 37

$$x-4 = \frac{x}{2} + 7 \quad \text{or} \quad -(x-4) = \frac{x}{2} + 7$$

$$2x-8 = x+14 \quad -2x+8 = x+14$$

$$x = 22$$

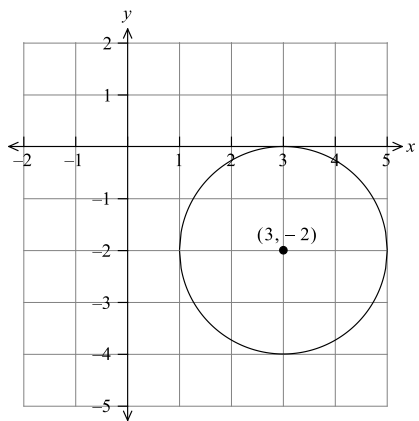
$$3x = -6$$

$$x = -2$$

$$\therefore x = 22 \text{ or } -2$$

23. Functions, 2ADV F1 2016 HSC 11a

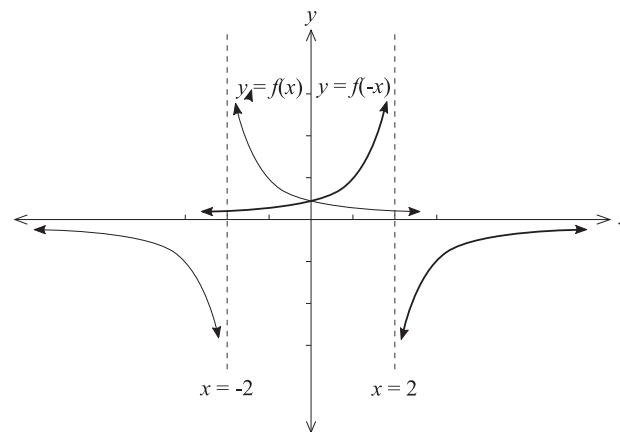
$(x-3)^2 + (y+2)^2 = 4$ is a circle,
centre $(3, -2)$, radius 2.



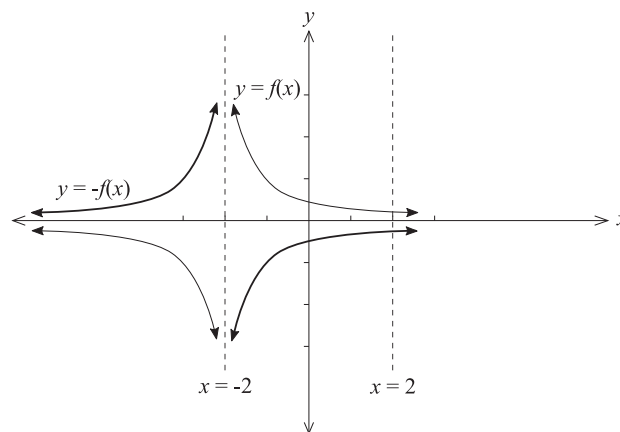
24. Functions, 2ADV F1 SM-Bank 36

i. Sketch $y = \frac{1}{x+2}$

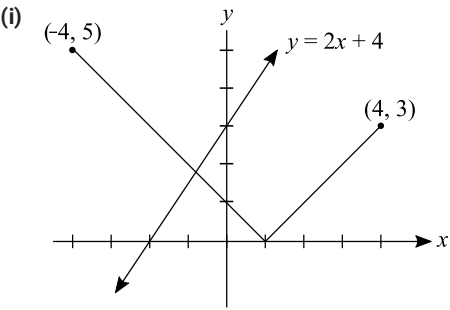
$y = f(-x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the y -axis.



ii. $y = -f(x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the x -axis.



25. Functions, 2ADV F1 2019 HSC 13e



ii. By inspection, intersection when $x = -1$

Test:

$$\begin{aligned}|-1-1| &= -2+4 \\ 2 &= 2\end{aligned}$$

$\therefore$ Intersection at $(-1, 2)$

26. Functions, 2ADV F1 EQ-Bank 45

$$f(x) = 2x^3 - 5x$$

Function is odd if: $f(-x) = -f(x)$

$$\begin{aligned}f(-x) &= 2(-x)^3 - 5(-x) \\ &= -2x^3 + 5x \\ &= -(2x^3 - 5x) \\ &= -f(x)\end{aligned}$$

$\therefore f(x)$ is odd.

27. Functions, 2ADV F1 EQ-Bank 6

Domain: $[-4, 2) \cup (2, \infty)$

Range: $(-\infty, \infty)$

28. Functions, 2ADV F1 EQ-Bank 7

$$A \propto \frac{1}{R} \Rightarrow A = \frac{k}{R}$$

When $A = 3$, $R = 28$:

$$3 = \frac{k}{28}$$

$$k = 84$$

Find A when $R = 8$:

$$A = \frac{84}{8} = 10.5$$

29. Functions, 2ADV F1 EQ-Bank 8

$$M \propto d^3$$

$$M = kd^3$$

When $M = 680$, $d = 15$

$$680 = k \times 15^3$$

$$k = 0.201481\dots$$

Find d when $M = 1100$:

$$1100 = 0.20148\dots \times d^3$$

$$d = \sqrt[3]{\frac{1100}{0.20148\dots}}$$

$$= 17.608\dots$$

$$= 17.6 \text{ cm (to 1 d.p.)}$$

30. Functions, 2ADV F1 EQ-Bank 26

Mass $\propto$ length³
 $m = kl^3$

Find k :

$$4350 = k \times 25^3$$
$$k = \frac{4350}{25^3}$$
$$= 0.2784$$

Find l when $m = 6200$:

$$6200 = 0.2784 \times l^3$$
$$l^3 = \frac{6200}{0.2784}$$
$$\therefore l = 28.13 \dots$$
$$= 28.1 \text{ cm (to 1 d.p.)}$$

31. Functions, 2ADV F1 EQ-Bank 27

Let d = stopping distance

$$d \propto s^2$$
$$d = ks^2$$

Find k ,

$$58.8 = k \times 70^2$$
$$k = \frac{58.8}{70^2}$$
$$= 0.012$$

Find d when $s = 105$:

$$d = 0.012 \times 105^2$$
$$= 132.3 \text{ metres}$$

32. Functions, 2ADV F1 SM-Bank 32

$$x^2 - 12x + y^2 + 2y - 12 = 0$$
$$(x - 6)^2 + (y + 1)^2 - 36 - 1 - 12 = 0$$
$$(x - 6)^2 + (y + 1)^2 = 49$$

$\therefore$ Centre $(6, -1)$

$\therefore$ Radius $= 7$

33. Functions, 2ADV F1 2010 HSC 1c

Circle with centre $(-1, 2)$, $r = 5$

$$(x + 1)^2 + (y - 2)^2 = 25$$

MARKER'S COMMENT: Expanding this equation is not necessary!

34. Functions, 2ADV F1 2010 HSC 1g

$$f(x) = \sqrt{x - 8}$$

Domain exists for:

$$(x - 8) \geq 0$$
$$x \geq 8$$

♦ Mean mark 49%.
MARKER'S COMMENT: $x > 8$ was a common incorrect answer.

35. Functions, 2ADV F1 2017 HSC 11h

Domain of $f(x) = \sqrt{3 - x}$

$$3 - x \geq 0$$
$$x \leq 3$$

Note domain can also be expressed as: $(-\infty, 3]$

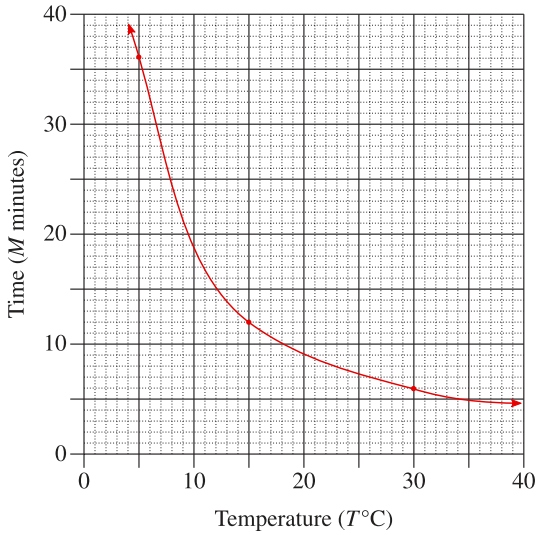
36. Functions, 2ADV F1 2022 HSC 12

a. $M \propto \frac{1}{T}$

$$M = \frac{k}{T}$$
$$12 = \frac{k}{15}$$
$$k = 15 \times 12$$
$$= 180$$
$$\therefore M = \frac{180}{T}$$

b.

T	5	15	30
M	36	12	6



♦ Mean mark (a) 49%.

37. Functions, 2ADV F1 2020 HSC 24

$$x^2 - 6x + y^2 + 4y - 3 = 0$$
$$x^2 - 6x + 9 + y^2 + 4y + 4 - 16 = 0$$
$$(x - 3)^2 + (y + 2)^2 = 16$$

$\Rightarrow$ Original circle has centre $(3, -2)$, radius $= 4$

Reflect in x -axis:

Centre $(3, -2) \rightarrow (3, 2)$

♦ Mean mark 48%.

